

EECE 455/632 - Cryptography and Network Security

Projects Ideas

Below is a set of simple projects that cover the classical encryption techniques as well as some basic tools needed in cryptography. Most of these ideas can be found implemented online. However, the idea is to package them with a nice graphical user interface (GUI) and make them user friendly and suitable for educational purposes.

Projects will be approved on first-come first-served basis. Each project will be assigned to a maximum of One group.

You can also propose your own Cryptography or Network Security project idea.

Implementation of Modern Techniques

Project-1. Classical Encryption

The following techniques should be implemented within the same platform.

1. Affine Cipher ($C = aP + b$): Given a and b , this code will allow the user to encrypt or decrypt a message whereby only characters are encrypted/decrypted and everything else remaining the same, such as a space, punctuation marks, newline, etc. Also, the code will give the user the option to crack an affine cipher technique by taking in the new most frequently used two letters (instead of E and T).
2. Mono-Alphabetic Cipher: Given a key consisting of 26 letters, the code will allow a user to encrypt or decrypt a message whereby only characters are encrypted/decrypted and everything else remaining the same, such as a space, punctuation marks, newline, etc.
3. Vigenère Cipher: Given a key that consists of a word or a sentence, the code will allow a user to encrypt or decrypt a message whereby only characters are encrypted/decrypted and everything else remaining the same, such as a space, punctuation marks, newline, etc.
4. Playfair Cipher: Given a key that consists of a word or a sentence, the code will display the Playfair Matrix and will allow a user to encrypt or decrypt a message whereby only characters are encrypted/decrypted and everything else remaining the same, such as a space, punctuation marks, newline, etc.
5. Hill Cipher: This code will support a matrix size of 2×2 or 3×3 . Given a particular matrix representing the key, the code will display the Matrix and its inverse, and it will allow a user to encrypt or decrypt a message whereby only characters are encrypted/decrypted and everything else remaining the same, such as a space, punctuation marks, newline, etc.
6. Extended Euclid Algorithm: This code will find the inverse of an integer modulo another integer.

Project-2. *Polynomial Arithmetic*

The code will allow a user to perform arithmetic on polynomials in $GF(2^m)$ with $(2 \leq m \leq 8)$ and using the following Irreducible Polynomials:

Degree 2: $x^2 + x + 1$

Degree 3: $x^3 + x^2 + 1$

Degree 4: $x^4 + x^3 + 1$

Degree 5: $x^5 + x^2 + 1$

Degree 6: $x^6 + x + 1$

Degree 7: $x^7 + x + 1$

Degree 8: $x^8 + x^4 + x^3 + x + 1$

The operations to be performed are modulo reduction, finding the inverse, addition, subtraction, multiplication, and division.

The software should allow inputs and display outputs in either Binary or Hexadecimal.

Project-3. *Polynomial Arithmetic (More Challenging; More Grades)*

The code will allow a user to perform arithmetic on polynomials in $GF(2^m)$ with $m = 163$. The operations to be performed are modulo reduction, finding the inverse, addition, subtraction, multiplication, and division.

The implementation should be modular and can be easily extended to other standard powers such as: 163, 233, 239, 283, 409, 571.

The software should allow inputs and display outputs in either Binary or Hexadecimal.

Project-4. *DES Implementation*

Implement the DES algorithm (64-bit data block and 64-bit key)

The software should be able to show the results for each round separately and the key expansion for each round separately. Within a round, the software should display the result after each operation.

The software should take inputs of the Message and Key in Hexadecimal and it should display the results in Hexadecimal.

Project-5. *AES Implementation*

Implement AES-128, AES-192, and AES-256.

The software should be able to show the results for each round separately and the key expansion for each round separately.

The software should take inputs of the Message and Key in Hexadecimal and it should display the results in Hexadecimal as a **State** of (4 by 4) byte Matrix.

Project-6. *Simple-Operations AES*

Implement AES-128 using lookup tables.

The software should be able to show the results for each round separately and the key expansion for each round separately.

The software should take inputs of the Message and Key in Hexadecimal and it should display the results in Hexadecimal as a **State** of (4 by 4) byte Matrix.

Project-7. *AES Again*

1. Given a key (128 bits) and a one data block, this implementation will have a GUI showing the step-by-step encryption of a single AES round pausing after the execution of each block in the round. The second part of this implementation will execute a full AES encryption or decryption showing the values of the key and the data after each round.
2. Avalanche: This code will take first a key of 128 bits and a data block and it will store the data values after each round of encryption. Then, it will prompt the user for either a new data block for the same key, or a new key for the same data block and it perform AES again showing the number of bit mismatch between the first encryption and the second after each round.
3. Key Expansion: Given a key of 128 bits, this code will generate the 44 key words needed for the 10 rounds of AES encryption. The code will display together each group of 4 words.

Project-8. *AES Again*

Implement AES in 5 modes: ECB, CBC, CFB, OFB, and Counter mode where the user will set the initial counter value (padding of last block should be done with zeroes along with count of pad size; Initial vectors should be set to zeroes).

Project-9. *Stream Cipher*

Implement RC4 and A5 for GSM encryption.

Project-10. *Number Theory*

Implement code to perform: Prime Factorization, Totient function computation, Miller Rabin Algorithm, Fast Exponentiation, and Chinese remainder Theorem whereby given $M = m_1 m_2 m_3 \dots m_k$, the code to compute the set of a_k given A , or it would compute A given the set of a_k .

Project-11. *El-Gamal*

Implement the algorithms for El-Gamal Encryption and Decryption.

Project-12. *Public Key Cryptography*

Implement RSA with key sizes of 1024 and 2048 bits.

Project-13. *Elliptic Curve Cryptography*

Given an Elliptic Curve $E_p(a, b)$, write code to find all points on the curve, add two points on the curve and multiply a point by a scalar.

Project-14. *Elliptic Curve Cryptography*

Implement ECC encryption/decryption based on a curve $E_p(a, b)$.

Project-15. *ECC Digital Signature*

Implement the Elliptic Curve Digital Signature Algorithm (**ECDSA**)

Project-16. *ECC D-H Key Exchange*

Implement the **Curve25519**, which is an elliptic curve offering 128 bits of security and designed for use with the elliptic curve Diffie–Hellman (**ECDH**) key agreement scheme.

Project-17. *Steganography (Message and Image)*

Steganography is the practice of concealing a message within another message or a physical object. In computing/electronic contexts, a computer file, message, image, or video is concealed within another file, message, image, or video. The word steganography comes from Greek steganographia, which combines the words steganós meaning "covered or concealed", and -graphia meaning "writing"

Steganography using message and image. The code should allow the user to add a message and the carrier (message or image) and perform steganography.

The project should include both a research on the methods used with each carrier type and second an implement the best of each.

Project-18. *Detecting Steganography (Message and Image)*

Detecting Steganography in messages and image. The code should allow the user to input a carrier and check for any hidden messages.

The project should include both a research on the detection methods used with each carrier type and an implement the best of each.

Project-19. *Steganography (Audio and Video)*

Steganography is the practice of concealing a message within another message or a physical object. In computing/electronic contexts, a computer file, message, image, or video is concealed within another file, message, image, or video. The word steganography comes from Greek steganographia, which combines the words steganós meaning "covered or concealed", and -graphia meaning "writing"

Steganography using video and audio. The code should allow the user to add a message and the carrier (video, or audio) and perform steganography.

The project should include both a research on the methods used with each carrier type and second an implement the best of each.

Project-20. Detecting Steganography

Detecting Steganography in video and audio. The code should allow the user to input a carrier and check for any hidden messages.

The project should include both a research on the detection methods used with each carrier type and an implement the best of each.